

G SARAN KASHYAP

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Career Objective

To be part of a reputed organization where I can apply my problem-solving skills, creativity, and technical knowledge to contribute to impactful projects, support organizational goals, and grow professionally.

Education

VNR Vignana Jyothi Institute of Engineering and Technology, B.Tech in Electronics and Instrumentation Engineering CGPA: 9.26/10	Sept 2023 – Present
B.Tech Minor in Artificial Intelligence and Machine Learning CGPA: 9.5/10	Aug 2025 – Present
Narayana Junior College – MPC 97.2%	– May 2023
Narayana High School CGPA: 10/10	– June 2021

Work Experience

Milieu Global Pvt. Ltd. , Embedded Engineer Intern	January 2026 - Present
- Developed an IoT-based multi-gas monitoring system using ESP32 with multi-node LoRa communication and GSM-based cloud connectivity. - Designed a 3-node LoRa architecture with a relay node to extend communication range and enable reliable data transmission in constrained environments. - Implemented MQTT-based data publishing over GSM (SIM800) for real-time remote monitoring using a cloud broker (HiveMQ). - Built a WiFi AP-based web dashboard for system monitoring and runtime configuration of gas thresholds and calibration parameters. - Implemented OTA-style configuration updates for real-time threshold and calibration management using persistent NVS (Preferences) storage. - Adapted a base TypeScript/React thermal imaging dashboard for a mine-monitoring gas detection device, implementing interpolation logic and optimizing the MQTT + Ngrok-based data pipeline to reduce latency from 20+ seconds to under 10 seconds. - Integrated multiple sensors using I2C, SPI, and ADC with I2C multiplexer-based channel selection. - Developed a threshold-based alert system with LED indicators, buzzer alerts, OLED display logging, and SMS notifications. - Performed hardware debugging and system optimization, resolving LoRa frequency mismatch and GSM power stability issues. - Assisted in PCB design and schematic development for multi-sensor integration using EasyEDA.	
Hasol Energy Pvt Ltd , Product Engineer Intern	June 2025 – September 2025
- Developed firmware for a dual-axis solar tracker with focus on PWM-based motor control, real-time monitoring, and performance optimization. - Wrote embedded software enabling seamless hardware–software integration and refined control algorithms based on field performance data analysis. - Analyzed performance data and identified optimization opportunities for improving tracking accuracy and system reliability. - Collaborated with a multidisciplinary team to design, validate, and field-test functional prototype models for deployment.	

Projects

Fall Detection & Smart Helmet System 🧐

- Designed a wearable fall detection and smart helmet system for rider safety and health monitoring
- Implemented fall detection algorithm using ADXL345 accelerometer with acceleration threshold and pattern analysis; monitored vital signs via MAX30100 pulse oximeter

- Integrated Twilio API to send real-time SMS alerts during emergencies
- Built a basic interface to display live sensor data for monitoring

IoT-Controlled Car 🚗

- Built a car prototype controlled via IoT using ESP32 and the Blynk mobile app.. Integrated Xbox controller to enable both mobile and joystick-based operation
- Demonstrated reliable wireless control for real-time operation
- Gained hands-on experience in hardware–software interfacing

Disease Prediction System 🏥

- Built a disease prediction model using **Logistic Regression** (97.6% accuracy) and **KNN** (92.8% accuracy), evaluated across 7 disease classes including dengue, malaria, and typhoid.
- Developed a web interface for real-time symptom input and prediction output.
- Used **Joblib** for model serialization and deployment.

Anti-lock Braking System (ABS) Simulation 🚗

- Developed a Bang-Bang controller based ABS simulation model in MATLAB Simulink
- Simulated tire dynamics including slip ratio and braking behavior
- Tested performance under different road conditions
- Analyzed braking performance and stopping distance

AI Vault | Privacy Management Chrome Extension 🛡️

- Engineered a **local-first Chrome extension** providing unified privacy controls and secure workspace management across major LLM platforms, including ChatGPT, Claude, and Gemini.
- Architected a scalable cross-platform solution utilizing a **platform adapter pattern** and **Shadow DOM** for strict UI isolation, preventing conflicts with native site structures.
- Implemented robust local security measures, integrating **SHA-256 cryptographic hashing** for PIN/password authentication and URL obfuscation to protect user metadata.
- Designed a localized data storage architecture using **IndexedDB**, alongside custom state-management logic for automated inactivity locking and stealth modes.

Technical Skills

Embedded & IoT: ESP32, STM32, Arduino, RTOS, Linux, Firmware development , PCB Design

Communication: UART, I2C, SPI, LoRa, RF, GSM ,MQTT,WIFI

Languages: C, Python, Embedded C , JavaScript

Web Development: HTML, CSS, Node.js, Express (Basic), MongoDB, RESTful APIs

AI/ML: Machine Learning algorithms, Feature Engineering, Model Evaluation

Tools: KiCad, EasyEDA, PlatformIO, ESP-IDF, STM32CubeIDE, Arduino IDE, VS Code, GitHub, MATLAB

Core Competencies: Data Structures and Algorithms, Problem-solving, Hardware Debugging, System Integration

Achievements and Participation

Co-Curricular

- Secured **3rd place** in National Science Day 2025 at VNRVJIET
- Secured **8th place** out of 50 teams in 24-Hour Hackathon “**Innovathon**”
- Participated in 24hr hackathon “**Emerge**”
- Attended IoT workshop “**Laser**” conducted by ISOI
- Won several national-level Olympiads during schooling

Extra-Curricular

- Assisted with stall setup and participant guidance at INCC 6.0
- Assisted AIRBOTS club in organizing events during Convergence 2K25

Certifications

NPTEL – Real Time Operating System

NPTEL – A Brief Introduction of Micro-Sensors

Maven Silicon – Embedded System Design using C Programming